

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $f(x) = \begin{cases} ax+1 & \text{if } x < 1 \\ 3 & \text{if } x = 1 \\ bx^2+1 & \text{if } x > 1 \end{cases}$. If $f(x)$ is continuous at $x = 1$ then $(a - b)$ is equal to-
- (A) 0 (B) 1 (C) 2 (D) 4

MCT142

2. For the function $f(x) = \begin{cases} \frac{1}{x+2^{\left(\frac{1}{x-2}\right)}}, & x \neq 2 \\ k, & x = 2 \end{cases}$ which of the following holds?
- (A) $k = 1/2$ and f is continuous at $x = 2$ (B) $k \neq 0, 1/2$ and f is continuous at $x = 2$
 (C) f can not be continuous at $x = 2$ (D) $k = 0$ and f is continuous at $x = 2$.

MCT143

3. The function $f(x) = \frac{4-x^2}{4x-x^3}$, is -
- (A) discontinuous at only one point in its domain.
 (B) discontinuous at two points in its domain.
 (C) discontinuous at three points in its domain.
 (D) continuous everywhere in its domain.

MCT144

4. If $f(x) = \begin{cases} -4\sin x + \cos x & \text{for } x \leq -\frac{\pi}{2} \\ a\sin x + b & \text{for } -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \cos x + 2 & \text{for } x \geq \frac{\pi}{2} \end{cases}$ is continuous then:
- (A) $a = -1, b = 3$ (B) $a = 1, b = -3$ (C) $a = 1, b = 3$ (D) $a = -1, b = -3$

MCT145

5. The function $f(x) = \begin{cases} \frac{1}{4}(3x^2+1) & -\infty < x \leq 1 \\ 5-4x & 1 < x < 4 \\ 4-x & 4 \leq x < \infty \end{cases}$ is -
- (A) continuous at $x = 1$ & $x = 4$
 (B) continuous at $x = 1$, discontinuous at $x = 4$
 (C) continuous at $x = 4$, discontinuous at $x = 1$
 (D) discontinuous at $x = 1$ & $x = 4$

MCT146

6. If $f(x) = \frac{x^2 - bx + 25}{x^2 - 7x + 10}$ for $x \neq 5$ and f is continuous at $x = 5$, then $f(5)$ has the value equal to-
- (A) 0 (B) 5 (C) 10 (D) 25

MCT147

7. If $f(x) = \frac{x - e^x + \cos 2x}{x^2}$, $x \neq 0$ is continuous at $x = 0$, then -
 (A) $f(0) = \frac{5}{2}$ (B) $[f(0)] = -2$
 (C) $\{f(0)\} = -0.5$ (D) $[f(0)] \cdot \{f(0)\} = -1.5$
 where $[.]$ and $\{.\}$ denotes greatest integer and fractional part function
MCT148
8. $y = f(x)$ is a continuous function such that its graph passes through $(a, 0)$.
 Then $\lim_{x \rightarrow a} \frac{\log_e(1 + 3f(x))}{2f(x)}$ is -
 (A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{2}{3}$
MCT149
9. In $[1, 3]$, the function $[x^2 + 1], [.]$ denoting the greatest integer function, is continuous -
 (A) for all x (B) for all x except at nine points
 (C) for all x except at seven points (D) for all x except at eight points
MCT150
10. Number of points of discontinuity of $f(x) = [2x^3 - 5]$ in $[1, 2)$, is equal to -
 (where $[x]$ denotes greatest integer less than or equal to x)
 (A) 14 (B) 13 (C) 10 (D) 8
MCT151
11. If $f(x)$ is continuous and $f\left(\frac{9}{2}\right) = \frac{2}{9}$, then the value of $\lim_{x \rightarrow 0} f\left(\frac{1 - \cos 3x}{x^2}\right)$ is -
 (A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) 0 (D) data insufficient
MCT152
12. f is a continuous function on the real line. Given that $x^2 + (f(x) - 2)x - \sqrt{3} \cdot f(x) + 2\sqrt{3} - 3 = 0$.
 Then the value of $f(\sqrt{3})$
 (A) can not be determined (B) is $2(1 - \sqrt{3})$
 (C) is zero (D) is $\frac{2(\sqrt{3} - 2)}{\sqrt{3}}$
MCT153
13. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :
 (A) all integers (B) all integers except 0 & 1
 (C) all integers except 0 (D) all integers except 1
MCT154
14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function $\forall x \in \mathbb{R}$ and $f(x) = 5 \forall x \in \text{irrational}$. Then the value of $f(3)$ is -
 (A) 1 (B) 2 (C) 5 (D) cannot determine
MCT155

15. The function $f(x) = [x] \cdot \cos \frac{2x-1}{2} \pi$, where $[\cdot]$ denotes the greatest integer function, is discontinuous at :-
 (A) all x (B) all integer points
 (C) no x (D) x which is not an integer
MCT156
16. Number of points of discontinuity of $f(x) = \left\{ \frac{x}{2} \right\} + \left[\frac{x}{3} \right]$ in $[0, 20]$ is -
 (where $[\cdot]$ & $\{ \cdot \}$ denote greatest integer function and fractional part function respectively)
 (A) 14 (B) 15 (C) 10 (D) 11
MCT157
17. Let $f : R \rightarrow Q$ be a continuous function such that $f(2) = 7$, then-
 (A) $f(x)$ is always an even function
 (B) $f(x)$ is always an odd function
 (C) nothing can be said about $f(x)$ being even or odd
 (D) $f(x)$ is a strictly increasing function
MCT158
18. Consider $f(x) = \begin{cases} \frac{\sin(2\pi \sec x)}{e^x - 1 - x} & x \neq 0 \text{ or } (2n+1)\frac{\pi}{2}, n \in I \\ k & x = 0 \end{cases}$
 Value of k for which $f(x)$ is continuous at $x = 0$, is -
 (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π
MCT159
19. Let $f(x) = \begin{cases} \frac{e^x - e^{\sin x}}{ax^3} & ; x < 0 \\ b & ; x = 0 \\ \frac{x}{\ln(1+4x)} & ; x > 0 \end{cases}$. If f is continuous at $x = 0$, then $(3a + 4b)$ is equal to -
 (A) -3 (B) 0 (C) 3 (D) 4
MCT160
20. The number of values of $x, x \in [-2, 3]$ where $f(x) = [x^2] \sin(\pi x)$ is discontinuous is
 (where $[\cdot]$ denotes greatest integer function) -
 (A) 19 (B) 8 (C) 6 (D) 0
MCT161

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $f(x) = \begin{cases} \sin^{-1}\left(\frac{a}{\ln x} - \frac{b}{x-1}\right) & x \neq 1 \\ \frac{\pi}{3} & x = 1 \end{cases}$ is continuous at $x = 1$, then-

(A) $a = b$ (B) $a = \sqrt{3}$ (C) $b = \sqrt{3}$ (D) $b = -\sqrt{3}$

MCT036

2. Let $f(x) = \begin{cases} \frac{\ln(1+x) - x + kx^2}{x^3} & ; x > 0 \\ \ell & ; x = 0 \\ \frac{\tan \frac{x}{3}}{x} & ; x < 0 \end{cases}$, then-

(A) $f(x)$ is continuous at $x = 0$ if $k = \frac{1}{2}, \ell = \frac{1}{3}$
 (B) $f(x)$ is discontinuous at $x = 0$ if $\ell \neq \frac{1}{3}$
 (C) $f(x)$ has irremovable discontinuity at $x = 0$ if $k \neq \frac{1}{2}$
 (D) $f(x)$ has removable discontinuity at $x = 0$ if $k = \frac{1}{2}, \ell \neq \frac{1}{3}$

MCT037

3. Which of the following function(s) is/are discontinuous at $x = 0$?

(A) $f(x) = \sin \frac{\pi}{2x}, x \neq 0$ and $f(0) = 1$ (B) $g(x) = x \sin\left(\frac{\pi}{x}\right), x \neq 0$ and $g(0) = \pi$
 (C) $h(x) = \frac{|x|}{x}, x \neq 0$ and $h(0) = 1$ (D) $k(x) = \frac{1}{1 + e^{\cot x}}, x \neq 0$ and $k(0) = 0$.

MCT038

4. A function $f(x)$ is defined as $f(x) = \frac{A \sin x + \sin 2x}{x^3}, (x \neq 0)$. If the function is continuous at $x = 0$, then -

(A) $A = -2$ (B) $f(0) = -1$ (C) $A = 1$ (D) $f(0) = 1$

MCT039

5. Which of the following function(s) can't be defined at $x = 0$ to make it continuous at $x = 0$?

(A) $f(x) = \frac{1}{1+2^x}$ (B) $f(x) = \tan^{-1} \frac{1}{x}$ (C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x} + 1}$ (D) $f(x) = \frac{1}{\ln|x|}$

MCT040

6. Which of the following function(s) can be defined continuously at $x = 0$?

(A) $f(x) = \frac{1 - \sec^2 2x}{4x^2}$ (B) $g(x) = \frac{\csc x - 1}{x \csc x}$ (where $\csc x = \operatorname{cosec} x$)
 (C) $h(x) = \frac{\sin 5x}{x}$ (D) $l(x) = (1 + 2x^2)^{\frac{1}{x^2}}$

MCT041

7. If f is defined on an interval $[a, b]$. Which of the following statement(s) is/are INCORRECT?
- (A) If $f(a)$ and $f(b)$, have opposite sign, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
 (B) If f is continuous on $[a, b]$, $f(a) < 0$ and $f(b) > 0$, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.
 (C) If f is continuous on $[a, b]$ and there is a point c in (a, b) such that $f(c) = 0$, then $f(a)$ and $f(b)$ have opposite sign.
 (D) If f has no zeroes on $[a, b]$, then $f(a)$ and $f(b)$ have the same sign.

MCT042

8. Which of the following functions can be defined at indicated point so that resulting function is continuous -

(A) $f(x) = \frac{x^2 - 2x - 8}{x + 2}$ at $x = -2$ (B) $f(x) = \frac{x - 7}{|x - 7|}$ at $x = 7$
 (C) $f(x) = \frac{x^3 + 64}{x + 4}$ at $x = -4$ (D) $f(x) = \frac{3 - \sqrt{x}}{9 - x}$ at $x = 9$

MCT043

9. In which of the following cases the given equations has atleast one root in the indicated interval ?
- (A) $x - \cos x = 0$ in $(0, \pi/2)$
 (B) $x + \sin x = 1$ in $(0, \pi/6)$
 (C) $\frac{a}{x-1} + \frac{b}{x-3} = 0, a, b > 0$ in $(1, 3)$
 (D) $f(x) - g(x) = 0$ in (a, b) where f and g are continuous on $[a, b]$ and $f(a) > g(a)$ and $f(b) < g(b)$.

MCT044

10. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x) = \frac{f(x)}{g(x)}$, where h is a rational function such that

- (a) Domain of $h(x)$ is $\mathbb{R} - \{-1\}$
 (b) $\lim_{x \rightarrow \infty} h(x) = \infty$ and
 (c) $\lim_{x \rightarrow -1} h(x) = \frac{1}{2}$.

Which of the following is/are True ?

(A) $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x)) = -\frac{39}{4}$ (B) $g(x) = 4(x + 1)$
 (C) $\lim_{x \rightarrow 0} (3h(x) + f(x) - 2g(x)) = \frac{39}{4}$ (D) $g(x) = 2(x + 1)$

MCT045

11. If the function $f(x) = \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$, ($x \neq -2$) is continuous at $x = -2$, then

(A) $a = 15$ (B) $a = -15$ (C) $f(-2) = 1$ (D) $f(-2) = -1$

MCT046

12. Let $f(x) = \begin{cases} |ax+3| & \text{if } x \leq -1 \\ |3x+a| & \text{if } -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & \text{if } 0 < x < \pi \\ \cos^2 x - 3 & \text{if } x \geq \pi \end{cases}$. If $f(x)$ is continuous for all $x \in \mathbb{R}$, then

(A) $a = 0$ (B) $a = 1$ (C) $b = 1$ (D) $b = -1$

MCT047

13. Let $f(x) = \begin{cases} \frac{\ln \cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x > 0 \\ \frac{e^{\sin 4x}-1}{\ln(1+\tan 2x)} & \text{if } x < 0 \end{cases}$
- (A) $f(0^+) = -2$ (B) $f(0^-) = 2$
 (C) $f(0^+) = 2$ (D) $f(0)$ can not be defined to remove discontinuity
- MCT048**

14. Let $f(x) = \frac{\sin^{-1}(1-\{x\}) \cdot \cos^{-1}(1-\{x\})}{\sqrt{2\{x\}} (1-\{x\})}$ then
 (where $\{.\}$ denotes the fractional part function)
- (A) $\lim_{x \rightarrow 0^+} f(x) = \frac{\pi}{2}$ (B) f is discontinuous at $x = 0$
 (C) $\lim_{x \rightarrow 0^-} f(x) = \frac{\pi}{2\sqrt{2}}$ (D) f is continuous at $x = 0$
- MCT049**

15. Let the function $f(x) = \begin{cases} ax-b & \text{for } x \leq 1 \\ 3x & \text{for } 1 < x < 2 \\ bx^2-a & \text{for } x \geq 2 \end{cases}$ is continuous at $x = 1$ but discontinuous at $x = 2$,
 then
- (A) $a - b = 3$ (B) $a - b = 6$ (C) $b \neq 3$ (D) $b = 3$
- MCT050**

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 16 to 18

Let $f(x) = \lim_{n \rightarrow \infty} \frac{\ln x + x^n}{(x-1) + x^n} \quad (x > 0)$.

On the basis of above information, answer the following questions:

16. $\lim_{x \rightarrow 1^-} \frac{\sin(f(x)-1)}{(x-1)}$ is equal to -
- (A) 0 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 1
- MCT053**

17. For $f(x)$ which of the following statement is true -
- (A) $f(x)$ is discontinuous at $x = 1$
 (B) non-negative difference between LHL & RHL at $x = 1$ is equal to 2.
 (C) $f(x)$ is many-one function
 (D) $f(x)$ is one-one function
- MCT054**

18. The number of solutions of the equation $f(x) = \frac{1}{(x-1)^2}$ is -
- (A) 0 (B) 1 (C) 2 (D) 3
- MCT055**

Paragraph for Question No. 19 to 20

Consider $f(x) = \frac{1}{\ln|x|}$ and $g(x) = \tan^{-1}|x|$.

On the basis of above information, answer the following questions :

19. Sum of solutions of the equation $\frac{g(x)}{f(x)} = \frac{1}{2}$ will be less than or equal to :

(A) 0 (B) 1 (C) 2 (D) 3

MCT062

20. The value(s) of x at which $h(x) = f(x) + g(x)$ will be discontinuous is/are :

(A) 0 (B) 1 (C) -1 (D) 2

MCT063

MATRIX MATCH TYPE QUESTION

21. **Column-I** **Column-II**

(A) $\lim_{x \rightarrow 1} \frac{x^3 - 1}{\ln x}$ is (P) 2

(B) $\lim_{x \rightarrow 0} \frac{x(\cos x - \cos 2x)}{2 \sin x - \sin 2x}$ is (Q) 3

(C) $\lim_{x \rightarrow 0} \frac{\tan x \sqrt{\tan x} - \sin x \sqrt{\sin x}}{x^3 \cdot \sqrt{x}}$ is (R) $\frac{3}{2}$

(D) If $f(x) = \cos(x \cos \frac{1}{x})$ and $g(x) = \frac{\ln(\sec^2 x)}{x \sin x}$ are (S) $\frac{3}{4}$

both continuous at $x = 0$ then $f(0) + g(0)$ equals

MCT064

MATCHING LIST TYPE QUESTION

22. Match the function in column-I with its behaviour at $x = 0$ in column-II, where $[.]$ denotes greatest integer function & $\operatorname{sgn}(x)$ denotes signum function.

Column-I	Column-II
(A) $f(x) = [x][1 + x]$	(P) LHL exist at $x = 0$
(B) $f(x) = [-x][1 + x]$	(Q) RHL exist at $x = 0$
(C) $f(x) = (\operatorname{sgn}(x))[2 - x][1 + x]$	(R) Continuous at $x = 0$
(D) $f(x) = [\cos x]$	(S) $\lim_{x \rightarrow 0} f(x)$ exists but function is discontinuous at $x = 0$
	(T) $\lim_{x \rightarrow 0} f(x)$ does not exist

MCT065

EXERCISE (S)

1. The function $f(x) = \begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0 < x < \frac{\pi}{2} \\ b+2 & \text{if } x = \frac{\pi}{2} \\ (1+|\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2} < x < \pi \end{cases}$

If $f(x)$ is continuous at $x = \pi/2$, then the value of $(a - b)$ is

MCT066

2. Let $f(x) = \begin{cases} \frac{1 - \sin \pi x}{1 + \cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases}$. The number of all possible values of 'p' so that the function

$f(x)$ is continuous at $x = 1/2$, is -

MCT067

3. Let $f(x) = \begin{cases} 1+x, & 0 \leq x \leq 2 \\ 3-x, & 2 < x \leq 3 \end{cases}$ and $g(x) = f[f(x)]$ & hence the number of points of discontinuity of $g(x)$ is

MCT068

4. If $f(x)$ is continuous at $x = \frac{\pi}{2}$ where $f(x) = \begin{cases} \frac{1 - \sin^3 x}{3\cos^2 x} & \text{if } x < \frac{\pi}{2} \\ a & \text{if } x = \frac{\pi}{2} \\ \frac{b(1 - \sin x)}{(\pi - 2x)^2} & \text{if } x > \frac{\pi}{2} \end{cases}$,

then the value of $(4a + 3b)$ is

MCT069

5. If the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x < 0 \\ c & \text{for } x = 0 \\ \frac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x > 0 \end{cases}$ is continuous at $x = 0$, then the value of

$(2a + 14c)$ is

MCT070

6. If $f(x) = \frac{\sin 3x + A \sin 2x + B \sin x}{x^5}$ ($x \neq 0$) is continuous at $x = 0$, then the value of $(A + B + f(0))$ is

MCT071

7. If $f(x) = x + \{-x\} + [x]$, where $[x]$ is the integral part & $\{x\}$ is the fractional part of x , then the number of points of discontinuity of $f(x)$ in $[-2, 2]$ is

MCT072

8. Let $f(x) = \begin{cases} (\sin x + \cos x)^{\csc x} & ; \quad -\frac{\pi}{2} < x < 0 \\ a & ; \quad x = 0 \\ \frac{e^{1/x} + e^{2/x} + e^{3/|x|}}{ae^{2/x} + be^{3/|x|}} & ; \quad 0 < x < \frac{\pi}{2} \end{cases}$

If $f(x)$ is continuous at $x = 0$, then the value of $(a^2 \cdot b^2)$ is

MCT073

9. Let f be a real valued continuous function on $\mathbb{R}$ and satisfying $f(-x) - f(x) = 0 \quad \forall \in \mathbb{R}$. If $f(-5) = 5$, $f(-2) = 4$, $f(3) = -2$ and $f(0) = 0$ then find the minimum number of zero's of the equation $f(x) = 0$.

MCT074

10. Find the number of points of discontinuity of the function $f(x) = [5x] + \{3x\}$ in $[0, 5]$ where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively.

MCT075

11. If $f(x) = \begin{cases} \frac{1 - \sin^3 x}{3 \cos^2 x} & , \quad x < \pi/2 \\ a & , \quad x = \pi/2 \\ \frac{b(1 - \sin x)}{(\pi - 2x)^2} & , \quad x > \pi/2 \end{cases}$ is continuous at $x = \frac{\pi}{2}$. Then the value of $2a + b$ is

MCT095

12. If $g(x) = (|x-1| + |4x-11|) [x^2 - 2x - 2]$, then find the number of point of discontinuity of $g(x)$ in

$$\left(\frac{1}{2}, \frac{5}{2}\right)$$

{where $[.]$ denotes GIF}

MCT096

EXERCISE - JEE (Main) PYQ

1. Let $f: R \rightarrow R$ be a function defined as : $f(x) = \begin{cases} 5, & \text{if } x \leq 1 \\ a+bx, & \text{if } 1 < x < 3 \\ b+5x, & \text{if } 3 \leq x < 5 \\ 30, & \text{if } x \geq 5 \end{cases}$. Then, f is :

[JEE (Main) 2019]

(1) continuous if $a = 5$ and $b = 5$ (2) continuous if $a = -5$ and $b = 10$ (3) continuous if $a = 0$ and $b = 5$ (4) not continuous for any values of a and b

MCT025

2. If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by $f(x) = \begin{cases} \frac{\sqrt{2} \cos x - 1}{\cot x - 1}, & x \neq \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$ is continuous, then k is equal

to

[JEE (Main) 2019]

(1) $\frac{1}{2}$

(2) 1

(3) $\frac{1}{\sqrt{2}}$

(4) 2

MCT026

3. If the function $f(x) = \begin{cases} a|\pi - x| + 1, & x \leq 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$ is continuous at $x = 5$, then the value of $a - b$ is :-

[JEE (Main) 2019]

(1) $\frac{2}{5-\pi}$ (2) $\frac{2}{\pi-5}$ (3) $\frac{2}{\pi+5}$ (4) $\frac{-2}{\pi+5}$

MCT027

4. Let $[t]$ denote the greatest integer $\leq t$ and $\lim_{x \rightarrow 0} x \left[\frac{4}{x} \right] = A$. Then the function, $f(x) = [x^2] \sin(\pi x)$ is discontinuous, when x is equal to :

[JEE (Main) 2020]

(1) $\sqrt{A+5}$ (2) $\sqrt{A+1}$ (3) $\sqrt{A}$ (4) $\sqrt{A+21}$

MCT028

5. Let $f(x) = x \cdot \left[\frac{x}{2} \right]$, for $-10 < x < 10$, where $[t]$ denotes the greatest integer function. Then the number of points of discontinuity of f is equal to.

[JEE (Main) 2020]

MCT029

6. Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be defined as

$$f(x) = \begin{cases} x+a, & x < 0 \\ |x-1|, & x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x+1, & x < 0 \\ (x-1)^2 + b, & x \geq 0 \end{cases}$$

where a, b are non-negative real numbers. If $(g \circ f)(x)$ is continuous for all $x \in R$, then $a + b$ is equal to.

[JEE (Main) 2021]

MCT030

7. If the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain and $f(0) = \frac{1}{k}$, then k is. [JEE (Main) 2021]

MCT031

8. Let a function $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined as

$$f(x) = \begin{cases} \sin x - e^x & \text{if } x \leq 0 \\ a + [-x] & \text{if } 0 < x < 1 \\ 2x - b & \text{if } x \geq 1 \end{cases}$$

Where $[x]$ is the greatest integer less than or equal to x . If f is continuous on $\mathbf{R}$, then $(a + b)$ is equal to: [JEE (Main) 2021]

- (1) 4 (2) 3 (3) 2 (4) 5d

MCT032

9. The number of points where the function

$$f(x) = \begin{cases} |2x^2 - 3x - 7| & \text{if } x \leq -1 \\ [4x^2 - 1] & \text{if } -1 < x < 1 \\ |x + 1| + |x - 2| & \text{if } x \geq 1 \end{cases}$$

$[t]$ denotes the greatest integer $\leq t$, is discontinuous is. [JEE (Main) 2022]

MCT033

10. Let $f(x) = [2x^2 + 1]$ and $g(x) = \begin{cases} 2x - 3, & x < 0 \\ 2x + 3, & x \geq 0 \end{cases}$, where $[t]$ is the greatest integer $\leq t$. Then, in the open interval $(-1, 1)$, the number of points where $f \circ g$ is discontinuous is equal to. [JEE (Main) 2022]

MCT034

11. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined as

$$f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x - 1], & 0 \leq x < 1 \\ b + [\sin(\pi x)], & 1 \leq x < 2 \\ [e^{-x}] - c, & x \geq 2 \end{cases}$$

where $a, b, c \in \mathbf{R}$ and $[t]$ denotes greatest integer less than or equal to t . Then, which of the following statements is true? [JEE (Main) 2022]

- (1) There exists $a, b, c \in \mathbf{R}$ such that f is continuous of $\mathbf{R}$.
 (2) If f is discontinuous at exactly one point, then $a + b + c = 1$.
 (3) If f is discontinuous at exactly one point, then $a + b + c \neq 1$.
 (4) f is discontinuous at atleast two points, for any values of a, b and c .

MCT035

12. If the function $f(x) = \begin{cases} \frac{\log_e(1 - x + x^2) + \log_e(1 + x + x^2)}{\sec x - \cos x}, & x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \{0\} \\ k, & x = 0 \end{cases}$

is continuous at $x = 0$, then k is equal to :

- (1) 1 (2) -1 (3) e (4) 0 [JEE (Main) 2022]

MCT076

13. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be functions defined by

$$f(x) = \begin{cases} [x] & , \quad x < 0 \\ |1-x| & , \quad x \geq 0 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} e^x - x & , \quad x < 0 \\ (x-1)^2 - 1 & , \quad x \geq 0 \end{cases}$$

where $[x]$ denote the greatest integer less than or equal to x . Then, the function fg is discontinuous at exactly : [JEE (Main) 2022]

- (1) one point (2) two points (3) three points (4) four points

MCT077

14. Let $f(x) = [x^2 - x] + |-x + [x]|$, where $x \in \mathbb{R}$ and $[t]$ denotes the greatest integer less than or equal to t . Then, f is [JEE (Main) 2023]

- (1) continuous at $x = 0$, but not continuous at $x = 1$
 (2) continuous at $x = 0$ and $x = 1$
 (3) not continuous at $x = 0$ and $x = 1$
 (4) continuous at $x = 1$, but not continuous at $x = 0$

MCT060

15. Let $[x]$ be the greatest integer $\leq x$. Then the number of points in the interval $(-2, 1)$, where the function $f(x) = |[x]| + \sqrt{x - [x]}$ is discontinuous is. [JEE (Main) 2023]

MCT061

16. If the function $f(x) = \begin{cases} \frac{72^x - 9^x - 8^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}} & , \quad x \neq 0 \\ a \log_e 2 \log_e 3 & , \quad x = 0 \end{cases}$ is continuous at $x = 0$, then the value of a^2 is equal to [JEE (Main) 2024]

- (1) 968 (2) 1152 (3) 746 (4) 1250

MCT114

17. Let $f: [-1, 2] \rightarrow \mathbb{R}$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where $[t]$ denotes the greatest integer less than or equal to t . The number of points, where f is not continuous, is : [JEE (Main) 2024]

- (1) 6 (2) 3 (3) 4 (4) 5

MCT115

18. Let $[t]$ denote the greatest integer less than or equal to t . Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \left[\frac{x}{2} + 3 \right] - [\sqrt{x}]$. Let S be the set of all points in the interval $[0, 8]$ at which f is not continuous. Then $\sum_{a \in S} a$ is equal to _____. [JEE (Main) 2024]

MCT116

19. Let $f: (0, \pi) \rightarrow R$ be a function given by

$$f(x) = \begin{cases} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0 < x < \frac{\pi}{2} \\ a - 8, & x = \frac{\pi}{2} \\ (1 + |\cot x|)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2} < x < \pi \end{cases}$$

Where $a, b \in Z$. If f is continuous at $x = \frac{\pi}{2}$, then $a^2 + b^2$ is equal to _____. [JEE (Main) 2024]

MCT117

20. Consider the function.

$$f(x) = \begin{cases} \frac{a(7x - 12 - x^2)}{b|x^2 - 7x + 12|}, & x < 3 \\ \frac{\sin(x-3)}{2^{x-[x]}}, & x > 3 \\ b, & x = 3 \end{cases}$$

Where $[x]$ denotes the greatest integer less than or equal to x . If S denotes the set of all ordered pairs (a, b) such that $f(x)$ is continuous at $x = 3$, then the number of elements in S is :

[JEE (Main) 2024]

- (1) 2 (2) Infinitely many (3) 4 (4) 1

MCT118

EXERCISE - JEE (Advanced) PYQ

1. For every integer n , let a_n and b_n be real numbers. Let function $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}, \text{ for all integers } n.$$

If f is continuous, then which of the following holds(s) for all n ?

[IIT JEE 2012]

- (A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

MCT057

2. For every pair of continuous function $f, g: [0, 1] \rightarrow \mathbb{R}$ such that

$$\max \{f(x) : x \in [0, 1]\} = \max \{g(x) : x \in [0, 1]\},$$

the correct statement(s) is(are) :

[JEE (Advanced) 2014]

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
(B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0, 1]$
(C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0, 1]$
(D) $(f(c))^2 = (g(c))^2$ for some $c \in [0, 1]$

MCT058

3. Let $[x]$ be the greatest integer less than or equal to x . Then, at which of the following point(s) the function $f(x) = x \cos(\pi(x + [x]))$ is discontinuous?

[JEE (Advanced) 2017]

- (A) $x = -1$ (B) $x = 0$ (C) $x = 2$ (D) $x = 1$

MCT059

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	A	B	A	D	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	C	C	C	A	C	C	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,B,C,D	A,B,C,D	A,B	A,B,C	A,C,D	A,C,D	A,C,D	A,B,C,D	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,D	A,C	A,B,D	A,B,C	A,C	C	C	C	A,B,C,D	A,B,C
Que.	21					22				
Ans.	(A→Q; B→R; C→S; D→P)					(A→P,Q,R; B→P,Q,T; C→P,Q,T; D→P,Q,S)				

EXERCISE (S)

1.	1	2.	0	3.	2	4.	14	5.	4
6.	2.00	7.	5.00	8.	1.00	9.	5.00	10.	30.00
11.	5	12.	2						

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	2	8	1	6	2	7	62
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	2	4	2	2	3	17	81	4

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	
Ans.	B,D	A,D	A,C,D	